

INTEGRATED CASE

Allied Food Products

Capital Budgeting and Cash Flow Estimation

11-16 AFTER SEEING SNAPPLE'S SUCCESS WITH NONCOLA SOFT DRINKS AND LEARNING OF COKE'S AND PEPSI'S INTEREST, ALLIED FOOD PRODUCTS HAS DECIDED TO CONSIDER AN EXPANSION OF ITS OWN IN THE FRUIT JUICE BUSINESS. THE PRODUCT BEING CONSIDERED IS FRESH LEMON JUICE. ASSUME THAT YOU WERE RECENTLY HIRED AS ASSISTANT TO THE DIRECTOR OF CAPITAL BUDGETING, AND YOU MUST EVALUATE THE NEW PROJECT.

THE LEMON JUICE WOULD BE PRODUCED IN AN UNUSED BUILDING ADJACENT TO ALLIED'S FORT MYERS PLANT; ALLIED OWNS THE BUILDING, WHICH IS FULLY DEPRECIATED. THE REQUIRED EQUIPMENT WOULD COST \$200,000, PLUS AN ADDITIONAL \$40,000 FOR SHIPPING AND INSTALLATION. IN ADDITION, INVENTORIES WOULD RISE BY \$25,000, WHILE ACCOUNTS PAYABLE WOULD GO UP BY \$5,000. ALL OF THESE COSTS WOULD BE INCURRED AT $t = 0$. BY A SPECIAL RULING, THE MACHINERY COULD BE DEPRECIATED UNDER THE MACRS SYSTEM AS 3-YEAR PROPERTY. 33%, 45%, 15%, 7%

THE PROJECT IS EXPECTED TO OPERATE FOR 4 YEARS, AT WHICH TIME IT WILL BE TERMINATED. THE CASH INFLOWS ARE ASSUMED TO BEGIN 1 YEAR AFTER THE PROJECT IS UNDERTAKEN, OR AT $t = 1$, AND TO CONTINUE OUT TO $t = 4$. AT THE END OF THE PROJECT'S LIFE ($t = 4$), THE EQUIPMENT IS EXPECTED TO HAVE A SALVAGE VALUE OF \$25,000.

UNIT SALES ARE EXPECTED TO TOTAL 100,000 CANS PER YEAR, AND THE EXPECTED SALES PRICE IS \$2.00 PER CAN. CASH OPERATING COSTS FOR THE PROJECT (TOTAL OPERATING COSTS ^{B/4} ~~LESS~~ DEPRECIATION) ARE EXPECTED TO TOTAL 60 PERCENT OF DOLLAR SALES. ALLIED'S TAX RATE IS 40 PERCENT, AND ITS WEIGHTED AVERAGE COST OF CAPITAL IS 10 PERCENT. TENTATIVELY, THE LEMON JUICE PROJECT IS ASSUMED TO BE OF EQUAL RISK TO ALLIED'S OTHER ASSETS.

YOU HAVE BEEN ASKED TO EVALUATE THE PROJECT AND TO MAKE A RECOMMENDATION AS TO WHETHER IT SHOULD BE ACCEPTED OR REJECTED. TO GUIDE YOU IN YOUR ANALYSIS, YOUR BOSS GAVE YOU THE FOLLOWING SET OF QUESTIONS.

TABLE IC11-1. ALLIED'S LEMON JUICE PROJECT
(TOTAL COST IN THOUSANDS)

END OF YEAR:

0 1 2 3 4

I. INVESTMENT OUTLAY					
EQUIPMENT COST	_____				
INSTALLATION	_____				
INCREASE IN INVENTORY	_____				
INCREASE IN ACCOUNTS PAYABLE	_____				
TOTAL NET INVESTMENT	_____				
II. OPERATING CASH FLOWS					
UNIT SALES (THOUSANDS)			100		
PRICE/UNIT		\$ 2.00	\$ 2.00		
TOTAL REVENUES					\$200.0
OPERATING COSTS,					
DEPRECIATION				36.0	16.8
OPERATING INCOME BEFORE TAXES				\$ 44.0	
TAXES ON OPERATING INCOME		0.3			25.3
OPERATING INCOME AFTER TAXES				\$ 26.4	
DEPRECIATION		79.2		36.0	
OPERATING CASH FLOW	\$ 0.0	\$ 79.7			\$ 54.7
III. TERMINAL YEAR CASH FLOWS					
RETURN OF NET WORKING CAPITAL					
SALVAGE VALUE					
TAX ON SALVAGE VALUE					
TOTAL TERMINATION CASH FLOWS					
IV. NET CASH FLOWS					
NET CASH FLOW	(\$260.0)				\$ 89.7
V. RESULTS					
NPV =					
IRR =					
MIRR =					
PAYBACK =					